

HEAVEN'S LIGHT IS OUR GUIDE



RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF MECHATRONICS ENGINEERING, 21Series

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Course Title: Control Systems Sessional

Experiment No: 07

Experiment Name: Designing of an LQG Controller for a Given System to Investigate the System Response and Stability

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Experiment Name: Designing of an LQG Controller for a Given System to Investigate the System Response and Stability.

Objectives:

1. To know about the LQG controller.
2. To know about the parameters and gain of LQG controller.
3. To know how to design the LQG controller in MATLAB editor for the given system.
4. To know how to design the LQG controller in Simulink for the given system.
5. To know the stability of the given system.

Theory:

In control theory, the linear–quadratic–Gaussian (LQG) control problem is one of the most fundamental optimal control problems. It concerns linear systems driven by additive white Gaussian noise. The problem is to determine an output feedback law that is optimal in the sense of minimizing the expected value of a quadratic cost Criterion. Output measurements are assumed to be corrupted by Gaussian noise and the initial state, likewise, is assumed to be a Gaussian random vector.

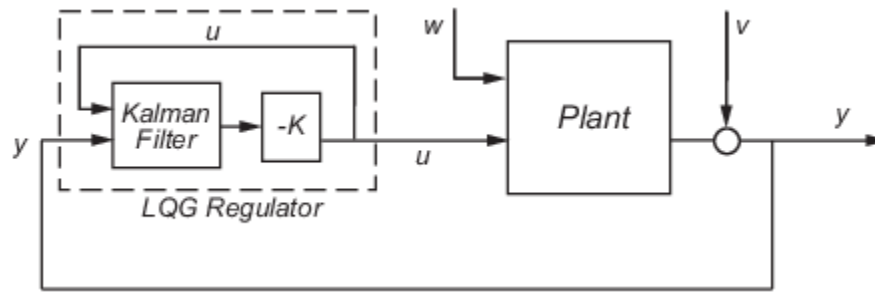


Figure 1: Basic block diagram of LQG Control System[1]

A possible generalization looks:

$$\begin{aligned}\dot{x} &= Ax + Bu + w_d \\ y &= Cx + w_n\end{aligned}$$

Where w_d , w_n are stochastic processes called process and measurement noise respectively. Only $y(t)$ is available for control. It turns out that for linear systems a separation principle holds.

First, calculating $\hat{x}(t)$ estimate the full state $x(t)$ using the available information and then, applying the LQR controller, using the estimation $\hat{x}(t)$ in place of the true state $x(t)$.

Defining the estimation error:

$$e = x - \hat{x}$$

The Kalman filter produces, $\hat{x}$ governed by an estimator dynamic equation:

$$\dot{\hat{x}} = A\hat{x} + Bu + K_f(y - C\hat{x})$$

Here K_f is the Kalman gain.

The control law is designed to stabilize the system: $u = -K\hat{x}$, where K is the LQR gain.

So, we can write

$$\dot{x} = Ax + Bu + w_d = Ax - BK\hat{x} + w_d$$

$$\dot{x} = Ax - BK\hat{x} - BKx + BKx + w_d = (A - BK)x + BKe + w_d$$

Differentiating $e = x - \hat{x}$, we get: $\dot{e} = \dot{x} - \dot{\hat{x}}$

As we have written,

$$\dot{\hat{x}} = A\hat{x} + Bu + K_f(y - C\hat{x}) = A\hat{x} - BK\hat{x} + K_f(y - C\hat{x})$$

Putting $y = Cx + w_n$ in the previous equation,

$$y - C\hat{x} = Cx + w_n - C\hat{x} = C[x - \hat{x}] + w_n = Ce + w_n$$

So, we can write,

$$\hat{\dot{x}} = A\hat{x} - BK\hat{x} + K_f(Ce + w_n) = \hat{\dot{x}} = A\hat{x} - BK\hat{x} + K_fCe + K_fw_n$$

Therefore,

$$\begin{aligned} \dot{e} &= \dot{x} - \hat{\dot{x}} = (Ax - BK\hat{x} + w_d) - (A\hat{x} - BK\hat{x} + K_fCe + K_fw_n) \\ \Rightarrow \dot{e} &= A(x - \hat{x}) - K_fCe + w_d - K_fw_n = (A - K_fC)e + w_d - K_fw_n \quad [2] \end{aligned}$$

Kalman Filter [2]

Kalman filtering is an algorithm that provides estimates of some unknown variables given the measurements observed over time. Kalman filters have been demonstrating its usefulness in various applications. Kalman filters have relatively simple form and require small computational power. Kalman Filter is an estimator for what is called the linear-quadratic problem, which is the problem of estimating the instantaneous “state” of a linear dynamic system perturbed by white noise-by using measurements linearly related to the state but corrupted by white noise.

Practically it is used for predicting the likely the future courses of dynamic systems that are likely to control.

$$\dot{x} = Fx + Gu + w$$

The measurements be linearly related

$$z = Hx + v$$

The process-noise matrix Q with noise vector w and measurement noise matrix R is related to the measurement noise vector v according to

$$Q = E[ww^T]$$

$$R = E[vv^T]$$

Parameter Covariance Matrix P [3]

‘P’ contains estimates of the uncertainty and correlation between uncertainties of state vector components. It is based on information supplied to the Kalman filter. It is not obtained from measurements.

Process Covariance Matrix Q(t)

Matrix describing instabilities of components of the state vector, e.g., clock noise, time transfer noise moving from time t to t+t.

Noise parameters:

Parameters used to determine the elements of the process covariance matrix, describe individual noise processes.

Measurement Covariance Matrix, R:

Describes measurement noise, white but individual measurements may be correlated. It may be removed.

Design Matrix, H:

Matrix that relates the measurement vector and the state vector using $y = Hx$.

Kalman Gain, K:

Kalman gain determines the weighting of current measurements and estimates from previous iteration. It is computed by filter.

Kalman gain computation

$$\begin{aligned} \dot{x} &= (A - BK)x + BKe + w_d \\ \dot{e} &= (A - K_fC)e + w_d - K_fw_n \end{aligned}$$

So, we can write,

$$\begin{bmatrix} \dot{x} \\ \dot{e} \end{bmatrix} = \begin{bmatrix} (A - BK) & BK \\ 0 & A - K_f C \end{bmatrix} \begin{bmatrix} x \\ e \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & -K_f \end{bmatrix} \begin{bmatrix} w_d \\ w_n \end{bmatrix}$$

where K_f is Kalman gain. It is expressed by

$$\underline{\underline{K}}(t_n) = \underline{\underline{P}}^-(t_n) \underline{\underline{H}}^T \left[\underline{\underline{H}} \underline{\underline{P}}^-(t_n) \underline{\underline{H}}^T + \underline{\underline{R}}(t_n) \right]^{-1}$$

Operation of Kalman Filter

1. It provides an estimate of the current parameters using current measurements and previous parameter estimates
2. It should provide a close to optimal estimate if the models used in the filter match the physical situation.

Five steps in the operation of a Kalman filter [4]

1. State Vector propagation
2. Parameter Covariance Matrix propagation
3. Compute Kalman Gain
4. State Vector update
5. Parameter Covariance Matrix update

The required code for designing LQG Controller for the given system:

The state space equation of the given system is

$$\begin{aligned} \dot{x} &= \begin{bmatrix} -1 & -5 \\ 10 & -1 \end{bmatrix} x + \begin{bmatrix} 10 \\ 0 \end{bmatrix} u \\ y &= [1 \ 0] x \end{aligned}$$

Code:

```
clc
clear all

A=[-1 -5; 10 -1];
B=[10; 0];
C=[1 0];
D=[0];

Qn=[1 0; 0 1];
Rn=0.1;
QXV = blkdiag([1 0; 0 1], 0.01);
QWV = blkdiag([1 0; 0 1], Rn);
QI=2e7*eye(output);

KLQG =lqg(sys,QXV,QWV,QI,'1dof');
CL =feedback(sys*KLQG, 1);
T= isstable(CL)

sys =ss(A,B,C,D);
state =2;
output =1;

figure(1)
step(sys)
hold on
step(CL,'r')
```

Output of Command Window

T =1

Result:

The open loop response of the system and close loop response of the system with LQG controller are:

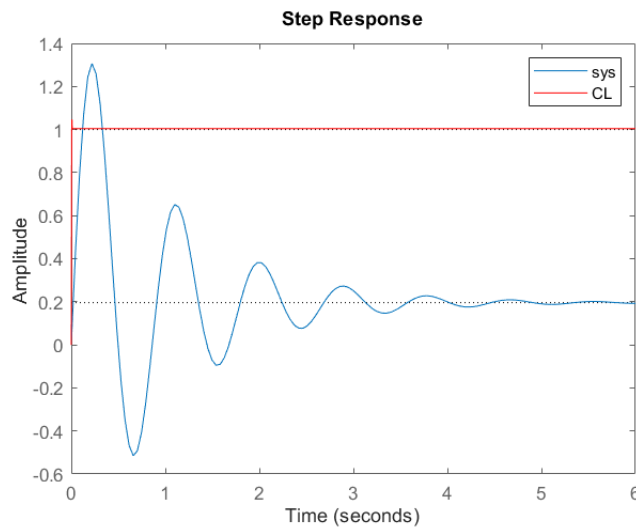


Figure 2: Open loop & closed loop step response of the system in MATLAB

Designing LQG Controller in Simulink:

The LQG controller for the given system is designed using block diagram in Simulink.

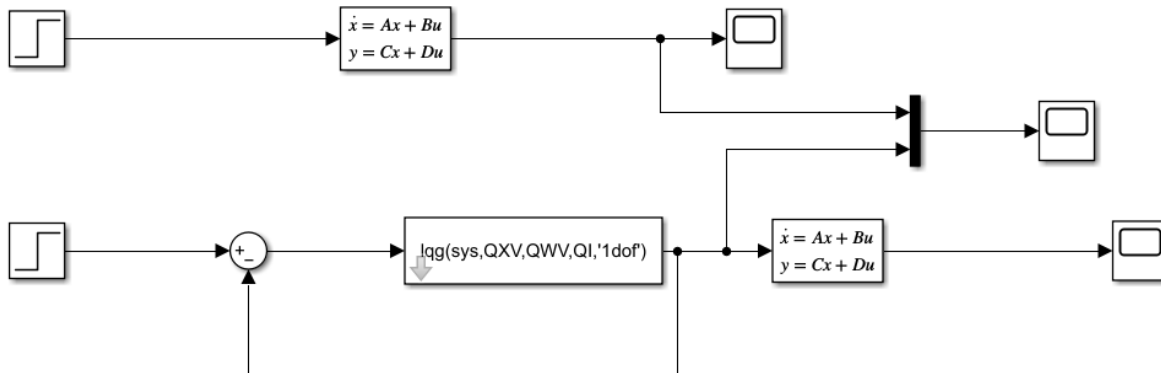


Figure 3: Open loop & closed loop Block Diagram the given system in Simulink

Result:

Using Simulink, the open loop response of the system and close loop response of the system with LQG controller are:

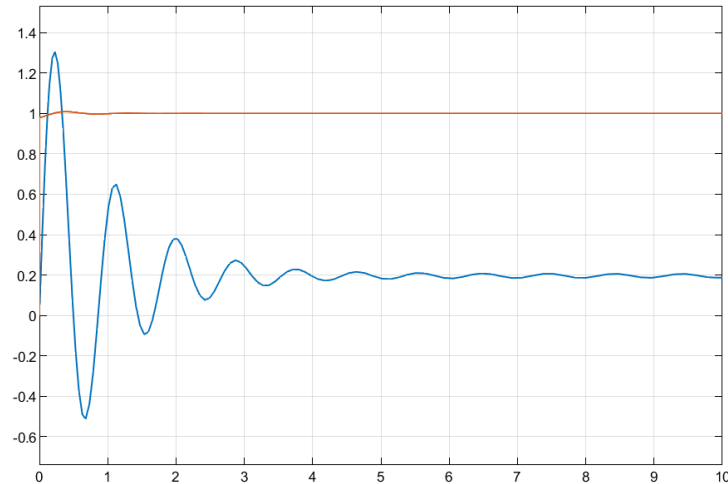


Figure 4: Open loop & closed loop step response simulation of the system in Simulink.

Discussion: In this experiment, a Linear Quadratic Gaussian (LQG) controller was designed and implemented for a given linear system using both MATLAB and Simulink. The LQG controller combines a Linear Quadratic Regulator (LQR) with a Kalman filter to provide optimal control under noisy conditions. The Kalman filter estimates the internal states from noisy outputs, while the LQR uses these estimates to generate the control input. The experiment demonstrated that the closed-loop system achieved better stability and performance compared to the open-loop response. Simulation results validated the theoretical expectations, showing an improved step response and confirming system stability ($T = 1$).

Conclusion: The experiment successfully demonstrated the design and implementation of an LQG controller for a linear system. By integrating the Kalman filter with LQR, the controller effectively estimated the system states and stabilized the system despite measurement noise. Both MATLAB and Simulink simulations confirmed that the closed-loop system provided improved dynamic performance and ensured system stability. This experiment highlights the practical usefulness of LQG control in optimal and robust system design.

Reference:

- [1] *Linear–Quadratic–Gaussian (LQG) Design*. MathWorks Documentation. Updated 2025. Available: <https://www.mathworks.com/help/control/getstart/linear-quadratic-gaussian-lqg-design.html>.
- [2] R. E. Kalman, “A new approach to linear filtering and prediction problems,” *Transactions of the ASME—Journal of Basic Engineering*, vol. 82, no. 1, Mar. 1960.
- [3] B. D. O. Anderson and J. B. Moore, *Optimal Filtering*. Englewood Cliffs, NJ, USA: Prentice-Hall, 1979.
- [4] K. J. Åström, *Introduction to Stochastic Control Theory*. Mineola, NY, USA: Dover Publications, 2006.